TREATMENT OF EPILEPSY THROUGH THE TRIGEMINAL NERVE: CLINICAL RESEARCH SUMMARY

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Abstract:

Multiple clinical experiences suggested that prolonged hyperactivity of the trigeminal nerve could precipitate a seizure. The purpose of this study was to see if reducing trigeminal nociceptive input, by way of orthopedic alignment of the mandible, could reduce seizures. Eight patients, who presented both seizures and cranio-mandibular dysfunction, were treated by jaw orthopedics so as speech and chewing were on the same trajectory. In all cases (except one in which jaw stability was never achieved), the patients either had no subsequent seizures, or a dramatically reduced seizure rate.

During my 20+ years experience, I have had many encounters with seizures that I suspected were related to dental procedures. Approximately three years ago, I had a patient who reported that she had not had a seizure in over a year, which paralleled the amount of time that she had been under my treatment for cranio-mandibular dysfunction. This was quite significant as she had previously had approximately six grand mal seizures a year, for over thirty years.

I then began to solicit and treat a number of patients with both seizures and cranio-mandibular dysfunction. I was curious whether this patient's response was just coincidental, unique, or whether her response could be replicated in other patients.

After eight cases of various types of seizures and three years experience, my data indicates that jaw orthopedics is extremely effective. In all cases, except one, there was a dramatic improvement in seizure rate and associated pathologies. In nearly all cases, there were no subsequent seizures once the jaw alignment was corrected and maintained.

Pathology:

Seizures are a common neurological disorder characterized by paroxymal, involuntary disturbances in brain function. They occur in 5.7% of the population. In over 70% of seizure patients, a cause is never found. Seizures can cause severe disruptions in quality of life. Of the patients taking medications, over 50% continue to have seizures, and over 50% have serious side effects from their medications.

Rationale:

Anatomy of Trigeminal Nerve:

Peripheral Anatomy: The trigeminal nerve is the largest nerve in the body with representation in 28% of the sensory cortex. Other than special sensory nerves (hearing, smell, sight, and taste), it is the principle sensory nerve for the head. Seldom is its broad range of sensory input recognized. Besides the more common known innervations of face, teeth pulp, oral cavity and gingiva, it also innervates the sinus membranes, front and back surface of the eye ball, anterior and middle meninges, and most of the scalp. It is unique in that it has an extremely high density of nerve endings. It is known to mediate the sense of smell, influence hearing and blood flow to the brain.

Central Nervous System Anatomy:

The three branches of the trigeminal nerve converge at the bilateral trigeminal ganglion where most sensory cell bodies are clustered, and then enters the brain stem at the mid pons region. The insertion location is such that it has a profound effect on the level of brain activity by way of its large representation and its direct connections with the reticular activation system. (It is these connections that you use to wake up in the morning by rubbing your eyes or washing your face.)

When the trigeminal nerve enters the brain stem, the neurons split, going in different directions depending on their type. The afferent fibers, which carry tactile information, are large diameter axioms that branch into two nuclei; principal sensory nuclei and spinal nuclei. Pain and temperature afferents descend to the trigeminal spinal nucleus which also receives input from cranial nerves VII, IX, X and spinal nerves C 1 to C 3. Proprioceptive neurons ascend to the mesencephalic nucleus where the cell bodies are located (only primary sensory cell bodies located within the brain). The cell bodies of the motor neurons are in the pons.

The trigeminal system of motor and sensory nuclei are direct rostral extensions of the sensory and motor systems in the spinal cord. The principal sensory nucleus and its major thalamic projections are

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both functionally and anatomically identical to the dorsal column-medial leminiscal system of the spinal cord.

Trigeminal Nerve Projections:

•Neurons of the spinal trigeminal nucleus (pain and temperature sensations) project to the ventral posterior medial and intralaminar nuclei of the thalamus. These axons also send collaterals to the reticular formation.

•Most fibers of the principal sensory nucleus (tactile information) travel to the contralateral ventral posterior nucleus of the thalamus, a decussating pathway that joins the ascending spinal dorsal column-medial lemniscus. These project to the somotosensory cortex.

•The mesencephalic trigeminal nucleus (proprioceptive, mechano receptors) consist of a column of primary sensory neurons that develop from the neural crest cells and can therefore be considered equivalent to a peripheral sensory ganglion. The majority of peripheral fibers of the mesencephalic nucleus appear to travel with the mandibular branch of the trigeminal nerve. This nucleus has a strong influence on posture and hence can affect spinal input.

Possible Trigeminal Nerve Influences on Seizure Activity:

The trigeminal nerve effects CNS function through a large quantity of neural connections, extensive projections, and a strategic input into the reticular activation system. The reticular activation system is an area in the brain stem that stimulates the brain, and not only keeps us conscious, but it can have a profound effect on brain activity level (e.g., hyperactivity and sleep disorders associated with TMD). With excessive input from trigeminal nociceptive and proprioceptive sensory nerves, the brain might be predisposed to seizures. Clinical experience indicates that it often takes a number of days of excessive input (conditioning) before a seizure is precipitated.

Neuro-Chemistry:

The trigeminal nerve is known to have a predominant influence on substance P levels in the cerebrospinal fluid. Substance P is known to depolarize nerve cells, effect smooth muscles (blood vessels), and act as a neurotransmitter in many regions of the brain. Elevated levels of Substance P have been shown to correspond with seizure activity in rats.

Neuro-Pharmacology:

Tegretol, is one of the primary drugs of choice for many types of seizures. It has two accepted clinical uses, Trigeminal Neuralgia and seizures. I find it surprising that no one has ever sought to discover why it works for both disorders.

Influence on Cerebrovascular System:

The trigeminal nerve innervates the Circle of Willis and has been shown to effect blood flow to the brain. This mechanism might account for the primary pathology found occasionally in severe epilepsy which is an atrophying of the amygdala/hypocampal region.

Methods:

After a medical history was taken, all patients were subjected to the diagnostics of 1. temporomandibular joint imaging, 2. kinesiographic examination of jaw motion, 3. models of teeth, and 4. intra-oral pictures. Precision Block Appliances[1] were fabricated to stabilize the mandible so as speech and biting were performed on or very near the same trajectory in the sagital and frontal planes. The rationale being that speech is a muscular phenomenon and biting is a skeletal phenomenon, and in order to eliminate any musculoskeletal discrepancy, these should be coincidental. These were worn at all times in the daytime, including eating. A Nocturnal Orthopedic Positioner[2] was fabricated to wear at night time.

The occlusal interfaces of appliances were modified over a 6-9 month period of time to correct for changes that occurred in functioning and resting positions. Final adjustments were done with ultra-low frequency TENS and monitored with surface electromyography.

Results:

Of the eight patients treated, four never had any subsequent seizures over an average 24 month follow up. Stability was never achieved in one patient due to excessive missing teeth. She continued to have petit mals. Three patients reported significant increase in stability with rare seizures during subsequent orthodontic stabilization when occlusal stability was lost due to tooth movement.

Discussion:

When orthopedic dysfunction occurs within cranio-mandibular apparatus, it elicits an intense barrage of nociceptive and proprioceptive input, as anyone who has ever experienced TMJ pain on TIC-Douloreux will attest. This nociceptive input can manifest

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as pain, not only locally, but frequently cause pain and dysfunction in other distribution regions of the same nerve (i.e. headaches, neck aches, sinus pain, etc.). This study would indicate that it can even cause major dysfunction within the CNS.

It is not clear whether a large number of seizures are caused primarily by conditioning of the CNS from trigeminal hyperactivity, or whether jaw orthopedics is just an effective therapy. If trigeminal hyperactivity is the cause, that would explain why no cause is found in such a large percentage of seizure patients (70%). Only a very small percentage of the medical and dental community recognize jaw dysfunction and/or trigeminal hyperactivity. The high degree of success in this study may also be attributable to the refined treatment methods.

Summary:

Jaw orthopedic therapy proved to be extremely effective in treatment of multiple types of epilepsy. The mechanism of action might possibly be by way of effecting brain chemistry (eg. Substance P levels) and/or the reticular activating system. Other research which stimulated the vargus nerve, and hence the RAS, has also shown some success in reducing seizures. It would be desirable to follow up this study with a larger sample, physician support, and better monitoring of CNS electrophysiology.

[1] Precision Appliance; similar to Twin Block Crozats but with multiple different design parameters.
[2] Nocturnal Orthopedic Positioner; similar to neutral bite Bionator but with different requirements on construction bite.